

Thisara Weerakoon

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[LinkedIn](#) | [GitHub](#) | [Medium](#) | [Portfolio](#)



ABOUT ME

Apache Contributor and open source enthusiast with hands-on experience in building clean and maintainable software. I am a continuous learner with a keen interest in backend engineering, machine learning and exploring the intersection of compilers and AI to build efficient systems.

EDUCATION

- **University of Moratuwa** June 2022 - Present
B.Sc. Eng (Hons) Computer Science and Engineering, specializing in Data Science and Engineering
Minor in Mathematics
 - **CGPA: 3.87/4.00** (6 semesters)
 - Dean's list in **all 6 semesters**
- **Rahula College, Matara** March 2007-Oct.2020
GCE Advanced Level - Physical Science Stream - Combined Mathematics, Physics, Chemistry
 - **Island Rank: 54**
 - Z-Score : 2.6945, 3As

WORKING EXPERIENCE

- **WSO2 Lanka Pvt. Ltd** Dec 2024 - May 2025
Software Engineering Intern - Micro Integrator Team
 - Implemented core components for the [Apache Synapse Go](#) initiative. Created comprehensive developer documentation and engaged the [Synapse Dev Community](#) to establish a clear path for future contributors ([GitHub](#)).
 - Developed a Ballerina connector to connect and interact with [Hubspot Meetings API](#).
 - **Technologies:** Go
- **University of Moratuwa** Jul 2025 - Nov 2025
Teaching Assistant - Data Science and Engineering Project (CS3501)
 - Mentored undergraduate groups for the Data Science and Engineering project.

OPEN SOURCE CONTRIBUTIONS

- **Apache Synapse-Go** (Contributor)
Stack: Go | Integration | Enterprise Service Bus
 - Listed as [Apache Synapse Contributor](#) (committership pending).
 - Learned how large scale software is architected and implemented from the beginning in a scalable manner. Developed technical communication skills by engaging in the [Synapse Dev Community Mailing List](#).
 - While re-implementing the Java-based Synapse functionalities in Golang, I encountered several practical challenges including [inter-process communication](#) and [dynamic package-level logging](#). Developed reusable experimental solutions for these and shared them with the Golang community through technical blogs.
- **C2FO/VFS** 
Stack: Go | Virtual File System
 - Contributed to improve [FTP/SFTP implementation](#) and [error handling](#).
 - Learned to write clean code that adheres to the existing coding standards of the maintained codebase.
- **Apache Dubbo-Go** 
Stack: Go | LLM
 - Contributed to improve [llm-sample](#) in dubbo-go-samples.

PROJECTS [\(Visit more projects\)](#)

- **Compiler Based Approach for Optimising LLM Inference on CPU Architectures** Final Year Project (Ongoing)
Stack: Triton | PyTorch | LLVM | C++
 - This final year project focuses on development of a compiler stack that integrates natively with Pytorch to accelerate LLM inference on CPUs, ensuring immediate compatibility for existing ML workflows. The solution aims to democratize LLM deployment for GPU-limited environments.
- **Agent Mesh - Orchestrating enterprise AI agents** Experimental Project 
Stack: Go | gRPC | A2A protocol | Google ADK | Docker
 - This golang framework enables the secure scaling of multi-agent systems from single-process monoliths to distributed microservices by providing intent-based discovery and distributed context propagation.

InnoLink - Secure Investment Marketplace
Stack: SpringBoot | React | PostgreSQL | LangChain4j | Azure | GitHub Actions

- Developed a secure investment marketplace connecting early-stage innovators with accredited investors. Integrated a RAG-based multi-agent system for automated feasibility analysis. ([Website](#)).

Personal Project

TBT-Former: Temporal Action Localization
Stack: PyTorch | Video Understanding | Computer Vision

- Proposed a novel Transformer architecture for action localisation establishing new state-of-the-art performance benchmarks on the THUMOS14 and EPIC-Kitchens datasets. Authored the research paper **TBT-Former** and released the [full code](#).

Research Project | 

Robust Identification Model Using Single-Channel ECG Data
Stack: Tensorflow | PyTorch | Neurokit | SciPy | Streamlit

- This research project addresses the challenge of making the State-of-the-art ECG-based person identification models noise-resilient. As a practical application of this research, developed an interactive web-based dashboard for arrhythmia and myocardial infarction detection using low quality ECG signals.

Research Project(Sem. 5)
[Product Demo](#) | 

CERTIFICATIONS

• Deep Learning Specialization (5 courses)	Coursera
• Specialization in Natural Language Processing (4 courses)	Coursera
• Generative AI with Large Language Models	Coursera
• Cloud Computing Concepts	Coursera
• Cloud Computing Applications: Part 1 and 2	Coursera
• WSO2 Certified Micro Integrator Practitioner - V4	WSO2

SKILLS

- **Programming Languages:** Go ,Python, Java
- **Libraries and Frameworks:** SpringBoot, Gin, PyTorch, Triton, LangChain4j
- **Technical Fields:** Fullstack development, Compilers for AI, Machine Learning

ACHIEVEMENTS

- **Global Rank 307 - Country Rank 15 - IEEEExtreme 18.0**
IEEE
- **Silver Medal - Sri Lankan Physics Olympiad (SLPhO) - 2019**
Institute of Physics, Sri Lanka
- **Silver Medal - Sri Lankan Junior Science Olympiad(SLJSO)-2016**
Institute of Physics, Sri Lanka

LEADERSHIP EXPERIENCE

• President <i>Mathematics Society - University of Moratuwa</i>	Nov 2023 - Dec 2024
• Manager <i>Data Analysis Team, AIESEC in Colombo South</i>	Jan 2023 - Aug 2023
• Event Chair - JamborIEEE 2022 <i>IEEE Sri Lanka Section</i>	Oct 2022 - Dec 2022

TEACHING EXPERIENCE

• Sri Lanka Olympiad Mathematics Foundation <i>Trainer</i> ◦ Trainer for the teams representing Sri Lanka for Thailand International Mathematics Olympiad 2022 and Hong Kong International Mathematics Olympiad 2022.	Dec 2022 - Apr 2023
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ADDITIONAL INFORMATION

Full Name: Weerakoon Thisara Rathnayaka
Languages: English (average fluency), Sinhala (native fluency)
Interests: Chess, Basketball, Astronomy, Science Fictions, Event Organizing, Blogging

REFERENCES

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